

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Steps Performed:

1. Shut down the computer completely.
2. Switched off the power supply and unplugged the power cable.
3. Pressed the power button for a few seconds to discharge any remaining power.
4. Opened the computer cabinet/laptop back cover.
5. Located the RAM slot(s) on the motherboard.
6. Released the retaining clips on both sides of the RAM module.
7. Carefully removed the RAM module.
8. Checked for dust or damage on the RAM and slot.
9. Reinserted the RAM module firmly until both clips locked into place.
10. Closed the cabinet, reconnected the power cable, and powered on the system.

Observations:

- The RAM module was seated properly.
- No physical damage or dust was observed.
- The computer booted successfully after reseating the RAM.
- No POST errors or beep codes were heard.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix

Hint: Think about display issues, driver checks, and physical connections

- 1 Check for display issues such as a blank screen, artifacts, or flickering.
- 2 Verify that the monitor cable is securely connected to the graphics card.
- 3 Ensure the GPU is properly seated in the PCIe slot and all power connectors are attached.
- 4 Install or update the graphics driver and restart the computer.
- 5 Test the GPU in another PCIe slot or another computer, or test with a known working GPU.

6 Monitor GPU temperature and fan operation.

7 Confirm the fix by verifying that the system boots normally and graphics function correctly without errors.

1. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Before Reconnecting:

2. The SATA SSD was disconnected.
3. BIOS did not detect the storage device.
4. The system displayed "**No Boot Device Found**" (or "**Boot Device Not Found**").

After Reconnecting:

5. Reconnected the SATA cable and power cable securely.
6. BIOS successfully detected the SSD.
7. The operating system booted normally without any errors.
8. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Component Checked First: RAM

Reason:

Faulty or improperly seated RAM is a common cause of random restarts, system crashes, and blue screen errors.

Diagnostic Steps:

9. Power off the computer and disconnect the power.
10. Remove and reseat the RAM modules.
11. Test each RAM module individually.
12. Try a different RAM slot if available.
13. Inspect the RAM contacts for dust or damage.
14. Boot the computer and observe whether the random restarts continue.

15.If the issue persists, replace the RAM with a known working module for confirmation.

Conclusion:

After reseating and testing the RAM, the system should be monitored to confirm whether the random restart issue has been resolved.
